123. H.C.F. of 132, 204 and 228.

$132 = 2 \times 2 \times 3 \times 11;$

$204 = 2 \times 2 \times 3 \times 17;$

$228 = 2 \times 2 \times 3 \times 19$

$\therefore$ HCF of 132, 204 and 228 is $2 \times 2 \times 3$ i.e. 12

$\therefore$ Required HCF = 12

Alternative method:

$$
\begin{array}{r|l}
132 & 204 \; (1 \\
 & 132 \\
\hline
72 & 132 \; (1 \\
 & 72 \\
\hline
60 & 72 \; (1 \\
 & 60 \\
\hline
12 & 60 \; (5 \\
 & 60 \\
\hline
 & \times
\end{array}
$$

Again HCF of 12 and 228

$$
\begin{array}{r|l|l}
12 & 228 & 19 \\
 & 12 & \\
\hline
 & 108 & \\
 & 108 & \\
\hline
 & \times &
\end{array}
$$

$\therefore$ Required HCF = 12

124. The given three numbers are $2a$, $5a$ and $7a$. LCM of $2a$, $5a$ and $7a = 2 \times 5 \times 7 \times a = 70a$

125. Product of two numbers = 1500, HCF = 10

$$\text{LCM} = \frac{\text{Product of two numbers}}{\text{Their HCF}}$$

$$= \frac{1500}{10} = 150$$

Required LCM is 150.

126. LCM of 8, 12 and 16 = 48

$$
\begin{array}{l|l}
2 & 8 - 12 - 16 \\
\hline
2 & 4 - 6 - 8 \\
\hline
2 & 2 - 3 - 4 \\
\hline
 & 1 - 3 - 2
\end{array}
$$

$2 \times 2 \times 2 \times 2 \times 3 = 48$

$\therefore$ Required number $= 48a + 3$

Which is divisible by 7.

$\therefore x = 48a + 3 = 7 \times 6a + 6a + 3$

$= (7 \times 6a)+(6a + 3)$ which is divisible by 7.

i.e. $6a + 3$ is divisible by 7.

When $a = 3$, $6a + 3 = 18 + 3 = 21$ which is divisible by 7.

$\therefore x = 48 \times 3 + 3 = 144 + 3 = 147$

127. Number of pair of positive integers whose sum is 99 and HCF is 9 is (9, 90; (18, 81); (27, 72); (36, 63) ; (45, 54).

128. Let the numbers are $3x$ and $4x$

So, HCF $= x$

$\because$ HCF $\times$ LCM = Product of numbers

$\Rightarrow x \times 120 = 3x \times 4x$

$\Rightarrow x \times 120 = 12x^2$

$\Rightarrow 120 = 12x$

$\Rightarrow x = 10$

$\therefore$ Numbers are 30 and 40.

$\therefore$ Sum of two numbers = 30 + 40 = 70

129. The greatest 4 digit number is 9999

The LCM of 12, 18, 21, 28 is 252

On dividing 9999 by 252 the remainder comes out to be 171

$\therefore$ Required number = 9999 − 171 = 9828

130.

$$
\begin{array}{l|l}
5 & 45 - 75 - 90 \\
\hline
3 & 9 - 15 - 18 \\
\hline
3 & 3 - 5 - 6 \\
\hline
 & 1 - 5 - 2
\end{array}
$$

$\Rightarrow 5 \times 3 \times 3 \times 5 \times 2 = 450$

LCM of 45, 75, 90 is 450

$\therefore$ Traffic lights will change simultaneously after 7 minutes 30 seconds i.e. at 7 : 27 : 45 hours.